DIPLOMA PROGRAMME PRACTICE · HL · SET A

Chemistry / Paper 1A

Question booklet · 40 marks · 120 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator and a clean Chemistry data booklet. Record choices online or on separate paper.

QUESTION 1 [1]

An element has isotopes of relative masses 63 and 65. An enriched sample has relative atomic mass 64.30. What fraction of its atoms have 36 neutrons if the atomic number is 29?

A 0.650
B 0.300
C 0.350
D 0.700

QUESTION 2 [1]

An element occurs as 75% ^{35}X and 25% ^{37}X. For randomly paired X_2^+ ions, what is the expected ratio of peak intensities at m/z 70, 72 and 74?

A 9:6:1
B 3:2:1
C 9:3:1
D 3:1:0

QUESTION 3 [1]

Successive ionization energies of a main-group atom are 578, 1817, 2745 and 11,577 kJ mol^{-1}. What follows about the ion formed after removal of its first three electrons?

A Its next electron is removed from a more tightly bound inner shell
B It must contain exactly three electrons
C It has gained a filled outer p sublevel by electron addition
D Its fourth electron is easier to remove than its third

QUESTION 4 [1]

Ground-state convergence frequencies for two atomic series are 1.60×10^{15} and 2.40×10^{15} Hz. Their thresholds correspond to ionization. What is the ratio of the longer limiting wavelength to the shorter, and which threshold needs more energy?

A 1.50; the 2.40×10^{15} Hz threshold
B 1.50; the 1.60×10^{15} Hz threshold
C 0.667; the 2.40×10^{15} Hz threshold
D 2.25; the 1.60×10^{15} Hz threshold

QUESTION 5 [1]

A 0.880 g compound containing only C, H and O yields 1.760 g CO_2 and 0.720 g H_2O on complete combustion. Use C=12.0, H=1.00, O=16.0. What is its empirical formula?

A C_4H_8O
B CH_2O
C C_2H_6O
D C_2H_4O

QUESTION 6 [1]

A fixed amount of ideal gas occupies 480 cm³ at 300 K and 96.0 kPa. It is compressed to 320 cm³ and cooled to 270 K. What is the final pressure?

A 57.6 kPa
B 160.0 kPa
C 86.4 kPa
D 129.6 kPa

QUESTION 7 [1]

At the same pressure, HF boils at a higher temperature than HCl despite its smaller molar mass. Which change absorbs most of the energy when HF boils?

A Breaking the H–F covalent bond in each molecule
B Overcoming intermolecular attractions including hydrogen bonds
C Separating pre-existing H^+ and F^- ions in the liquid
D Removing an electron from fluorine

QUESTION 8 [1]

In one contributor to O_3, central O has one lone pair, one double bond and one single bond. Which formal charge assignment is correct in the order central, double-bonded terminal, single-bonded terminal?

A 0, 0, 0
B +1, 0, −1
C 0, +1, −1
D −1, 0, +1

QUESTION 9 [1]

ClF_3 and BrF_5 each have a halogen at the centre. Which pair gives their molecular geometries in that order?

A T-shaped and octahedral
B Trigonal planar and trigonal bipyramidal
C T-shaped and square pyramidal
D Trigonal pyramidal and square planar

QUESTION 10 [1]

How many sigma and pi bonds are present in CH_2=CH–C≡N, counting all C–H bonds?

A 5 sigma, 3 pi
B 7 sigma, 2 pi
C 6 sigma, 2 pi
D 6 sigma, 3 pi

QUESTION 11 [1]

A metal wire and its dilute substitutional alloy both conduct. The alloy resists bending more strongly. Which model accounts for both observations?

A Mobile electrons persist, while size differences hinder layers slipping
B Ions in the alloy move freely through the solid
C Every alloy atom forms four localized bonds
D Alloying removes the positive ions from the lattice

QUESTION 12 [1]

A linear chain forms from equimolar ethane-1,2-diol and a dicarboxylic acid. Joining six diol and six diacid molecules into one open chain eliminates how many water molecules? Assume every link is an ester.

A 12
B 10
C 11
D 6

QUESTION 13 [1]

Four species have ten electrons. Which has both the largest nuclear charge and the smallest ionic radius?

A Na^+
B O^{2-}
C F^-
D Al^{3+}

QUESTION 14 [1]

Two octahedral complexes absorb at 480 and 600 nm. Under the d-level splitting model, which comparison of gaps is correct?

A The 600 nm complex has a gap 1.25 times as large
B The 480 nm complex has a gap 1.25 times as large
C The 480 nm complex has a gap 2.25 times as large
D Both gaps are equal because the coordination numbers match

QUESTION 15 [1]

Which pair consists of structural isomers that belong to different homologous series?

A Propan-1-ol and methoxyethane
B Propene and propane
C Ethanoic acid and ethanol
D Butan-1-ol and butan-2-ol

QUESTION 16 [1]

A $C_4H_8O_2$ ester gives a 3H singlet near 3.7 ppm, a 2H quartet near 2.3 ppm and a 3H triplet near 1.1 ppm. Which structure fits?

A $CH_3COOCH_2CH_3$
B $HCOOCH(CH_3)_2$
C $CH_3CH_2COOCH_3$
D $CH_3CH_2CH_2COOH$

QUESTION 17 [1]

A compound has a molecular ion at m/z 74, a broad O–H absorption typical of an alcohol, and one 9H singlet plus an exchangeable 1H signal. Which structure fits?

A Butan-2-ol
B 2-methylpropan-2-ol
C Ethoxyethane
D Butan-1-ol

QUESTION 18 [1]

Which sample can show optical rotation under standard measurement conditions?

A An enantiopure sample of 2-bromobutane
B Pure butan-2-one
C Pure 2-bromopropane
D A 1:1 mixture of 2-bromobutane enantiomers

QUESTION 19 [1]

Two 40.0 cm^3 solutions, initially at the same temperature, react to give a temperature rise of 3.00 K. Assume density 1.00 g cm^{-3} and c=4.18 J g^{-1} K^{-1}. If 0.0100 mol reacts, what is the molar enthalpy change?

A +100 kJ mol^{-1}
B −50.2 kJ mol^{-1}
C −100 kJ mol^{-1}
D −1.00 kJ mol^{-1}

QUESTION 20 [1]

Given X→Y, ΔH=−42 kJ mol⁻¹, and Z→Y, ΔH=+17 kJ mol⁻¹, what is ΔH for X→Z on the same molar basis?

A −25 kJ mol⁻¹
B +59 kJ mol⁻¹
C −59 kJ mol⁻¹
D +25 kJ mol⁻¹

QUESTION 21 [1]

For formation of MX(s) from the elements, ΔHf°=−320 kJ mol⁻¹. The steps forming gaseous M^+ and X^- from the elements sum to +410 kJ mol⁻¹. What is the lattice dissociation enthalpy?

A +730 kJ mol⁻¹
B −730 kJ mol⁻¹
C +90 kJ mol⁻¹
D −90 kJ mol⁻¹

QUESTION 22 [1]

For ethanol(l), combustion to CO_2(g) and H_2O(l) has ΔH°=−1367 kJ mol⁻¹. Use ΔHf°(CO_2)=−394 and ΔHf°(H_2O(l))=−286 kJ mol⁻¹. What is ΔHf° of ethanol(l)?

A −1646 kJ mol⁻¹
B +279 kJ mol⁻¹
C −3013 kJ mol⁻¹
D −279 kJ mol⁻¹

QUESTION 23 [1]

At constant pressure, ΔH°=+36.0 kJ mol⁻¹ and ΔS°=+120 J K⁻¹ mol⁻¹. Assuming both constant, when is the forward reaction spontaneous under standard conditions?

A Only at exactly 300 K
B Below 300 K
C Above 300 K
D At every positive temperature

QUESTION 24 [1]

At 300 K a reaction has K=0.040. For a mixture with Q=0.010, which pair of signs is correct for ΔG° and the actual forward ΔG?

A Positive and positive
B Negative and negative
C Positive and negative
D Negative and positive

QUESTION 25 [1]

A sealed vessel contains 0.250 mol propane and 0.900 mol O_2. For complete combustion to CO_2 and H_2O, what is the greatest amount of CO_2 possible?

A 0.750 mol
B 0.900 mol
C 0.540 mol
D 0.180 mol

QUESTION 26 [1]

Identical excess acid is added to equal masses of pure carbonate as powder or chips. Which pair describes the powder relative to the chips if all CO_2 is collected at the same T and p?

A Same initial slope, greater final volume
B Same initial slope, smaller final volume
C Greater initial slope, greater final volume
D Greater initial slope, same final volume

QUESTION 27 [1]

Doubling [A] alone quadruples rate. Doubling [A] and tripling [B] together multiplies rate by 12. What are the overall order and the units of k for rate in mol dm^{-3} s^{-1}?

A Second order; dm^3 mol^{-1} s^{-1}
B Third order; dm^6 mol^{-2} s^{-1}
C Third order; dm^3 mol^{-1} s^{-1}
D Fourth order; dm^9 mol^{-3} s^{-1}

QUESTION 28 [1]

A fast pre-equilibrium $A+B \rightleftharpoons I$ is followed by slow $I+B \rightarrow P$. If I is an intermediate, what rate law follows?

A $rate=k_{obs}[A]^2[B]$
B $rate=k_{obs}[A][B]$
C $rate=k_{obs}[I][A]$
D $rate=k_{obs}[A][B]^2$

QUESTION 29 [1]

In a plot of ln k against 1/T, the gradient is −5700 K. What is Ea, using R=8.31 J K^{-1} mol^{-1}?

A −47.4 kJ mol^{-1}
B 0.686 kJ mol^{-1}
C 5.70 kJ mol^{-1}
D 47.4 kJ mol^{-1}

QUESTION 30 [1]

At fixed temperature a gaseous equilibrium $2NO_2 \rightleftharpoons N_2O_4$ is compressed. Which combination describes the new equilibrium?

A More N_2O_4 in moles; K increases

B No change in composition; K decreases

C More NO_2 in moles; K increases

D More N_2O_4 in moles; K unchanged

QUESTION 31 [1]

In a 1.00 dm^3 vessel $H_2+I_2 \rightleftharpoons 2HI$ starts with 0.500 mol each of H_2 and I_2 and no HI. At equilibrium 0.600 mol HI is present. What is Kc?

A 9.00

B 2.25

C 0.111

D 3.00

QUESTION 32 [1]

For $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, a mixture initially at equilibrium is instantaneously diluted to double its volume at fixed T. Immediately after dilution, how does Qc compare with Kc?

A Qc=Kc/4 and no shift occurs

B Qc=Kc and no shift occurs

C Qc=Kc/2 and the system shifts forward

D Qc=2Kc and the system shifts forward

QUESTION 33 [1]

A buffer contains 0.0600 mol HA and 0.0200 mol A^-. Add 0.0100 mol NaOH, with pKa=4.80. What is the resulting pH?

A 4.80

B 4.58

C 4.32

D 5.02

QUESTION 34 [1]

A weak acid of concentration 0.100 mol dm^{-3} has pH 2.50. Which estimate of Ka is closest, taking account of acid consumed in ionization?

A 1.00×10^{-5}

B 1.03×10^{-4}

C 3.27×10^{-2}

D 3.16×10^{-3}

QUESTION 35 [1]

A galvanic cell operates by $Mg(s)+Cu^{2+}(aq)\rightarrow Mg^{2+}(aq)+Cu(s)$. Which simultaneous changes occur?

A Cu electrode loses mass and electrons flow toward Mg

B Cu^{2+} is oxidized and anions travel through the wire

C Mg^{2+} is reduced and the Mg electrode grows

D Mg electrode loses mass and electrons flow externally toward Cu

QUESTION 36 [1]

Given $E^\circ(Ag^+/Ag)=+0.80$ V and $E^\circ(Zn^{2+}/Zn)=-0.76$ V, what are E°_{cell} and ΔG° for a cell reaction consuming 1 mol Zn? $F=96{,}500$ C mol^{-1}.

A +0.04 V and −7.72 kJ mol^{-1}

B +1.56 V and −301 kJ mol^{-1}

C +1.56 V and −151 kJ mol^{-1}

D −1.56 V and +301 kJ mol^{-1}

QUESTION 37 [1]

A metal couple has $E^\circ(M^{3+}/M^{2+})=+0.40$ V. Which oxidizing agent can oxidize M^{2+} to M^{3+} under standard conditions? Its reduction potential is supplied.

A Y_2, $E^\circ(Y_2/Y^-)=+0.10$ V

B X_2, $E^\circ(X_2/X^-)=+0.65$ V

C Z^+, $E^\circ(Z^+/Z)=-0.15$ V

D W^{2+}, $E^\circ(W^{2+}/W)=+0.35$ V

QUESTION 38 [1]

Which step both consumes a radical and produces a new radical in chlorination of ethane?

A $Cl\bullet + C_2H_6 \rightarrow HCl + C_2H_5\bullet$

B $C_2H_5\bullet + Cl\bullet \rightarrow C_2H_5Cl$

C $Cl_2 \rightarrow 2Cl\bullet$

D $2C_2H_5\bullet \rightarrow C_4H_{10}$

QUESTION 39 [1]

A primary bromoalkane undergoes a concerted substitution by OH^-. Which description of the transition state is correct?

A A planar carbocation forms before attack

B C–O formation and C–Br breaking occur together

C A stable intermediate has five complete bonds to carbon

D Br^- leaves after a free radical forms

QUESTION 40 **[1]**

Propene reacts with HBr by ionic electrophilic addition, without peroxides. Why is 2-bromopropane the main product?

A Bromide first removes H^+ from the alkene

B HBr adds through homolytic cleavage only

C Protonation leading to a secondary carbocation is favoured over the primary alternative

D The main product must be chiral